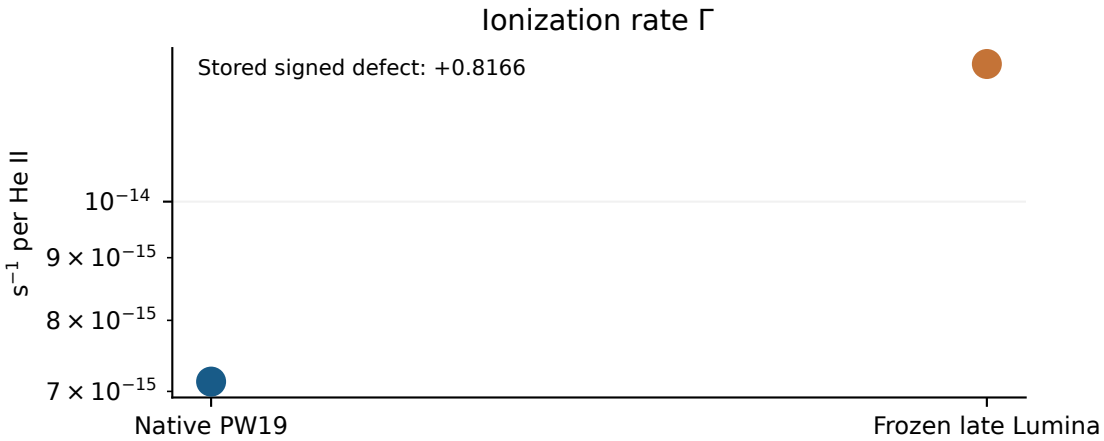
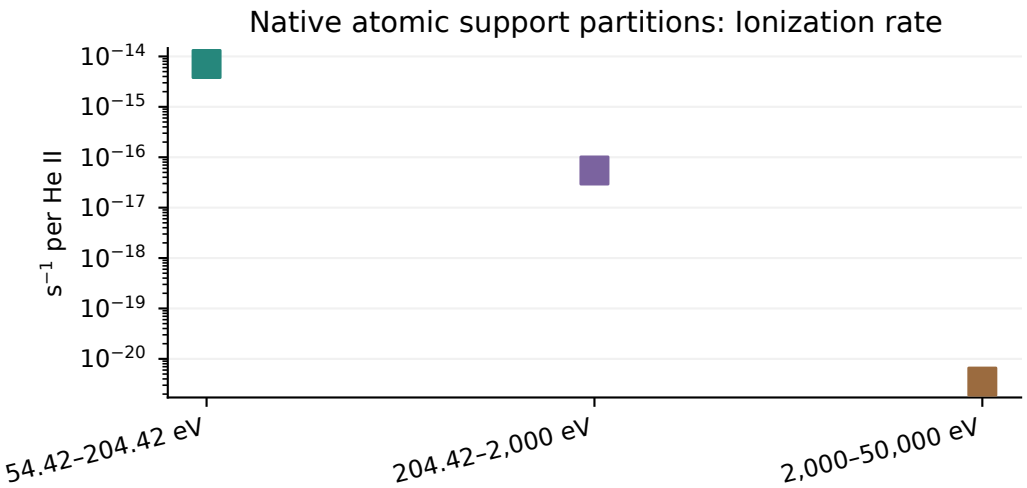
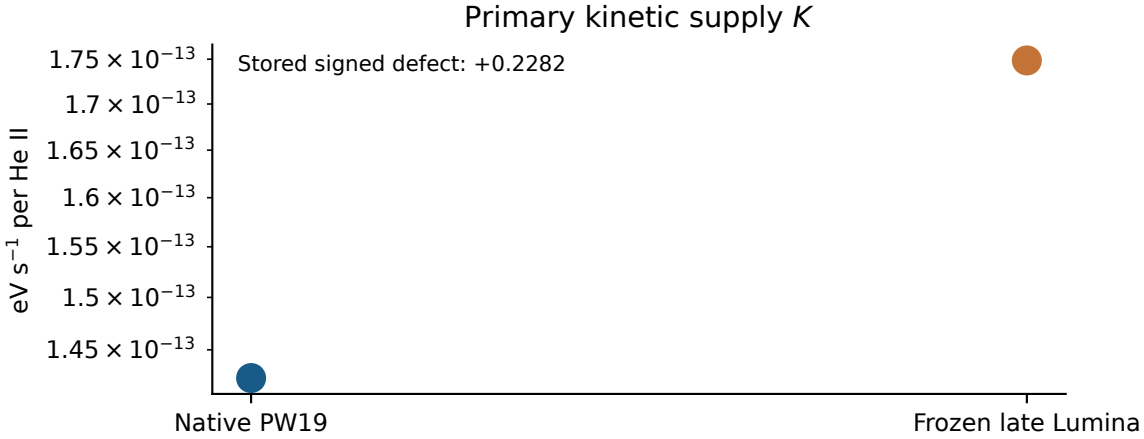
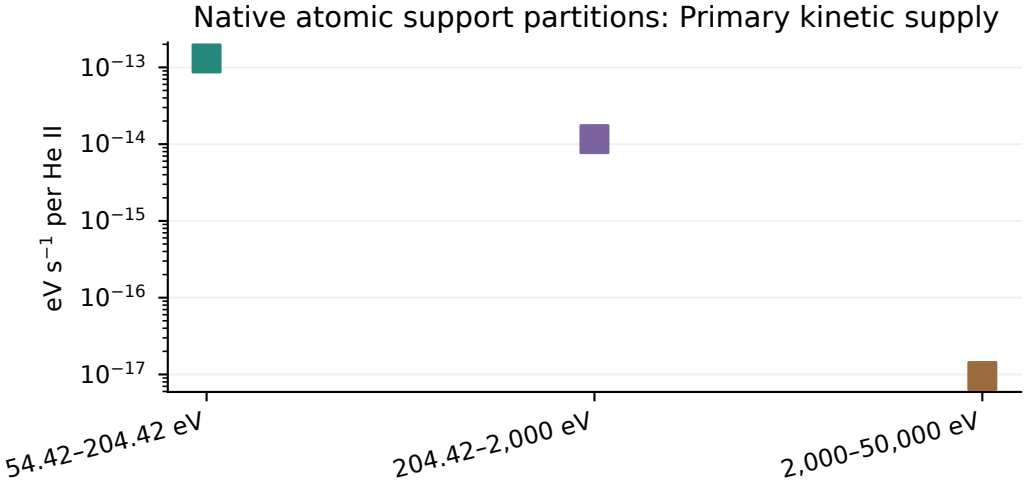
PW19 native He II source vs frozen late Lumina moment

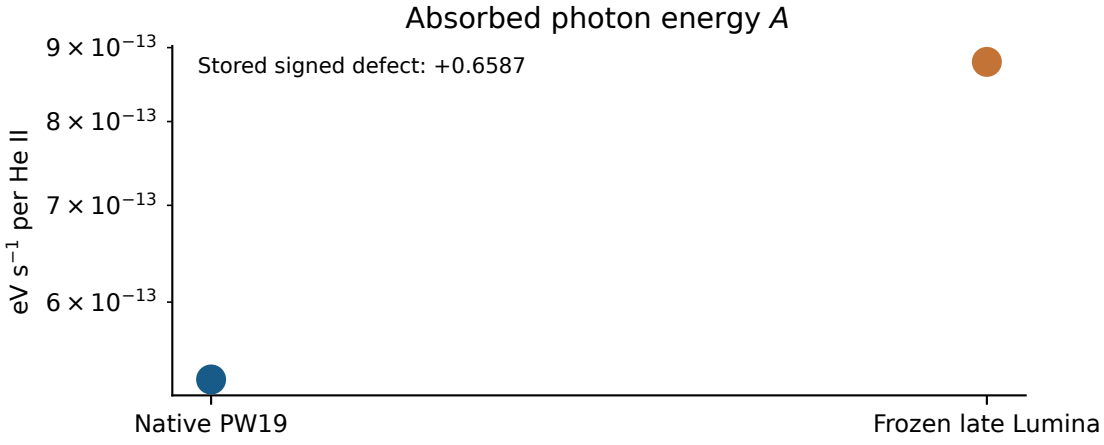
Exact source node z = 2.479 · same native late-band photon exposure · proper-second, bath-rest-frame rates

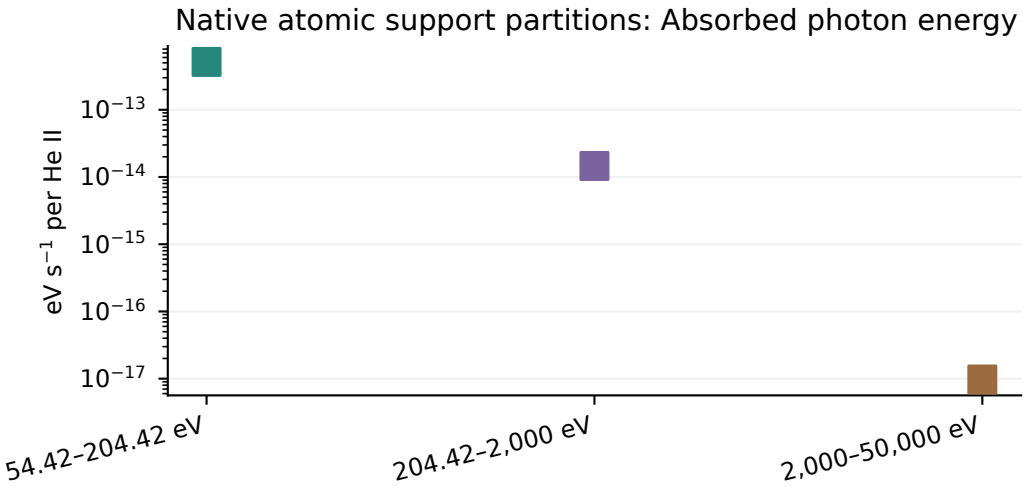


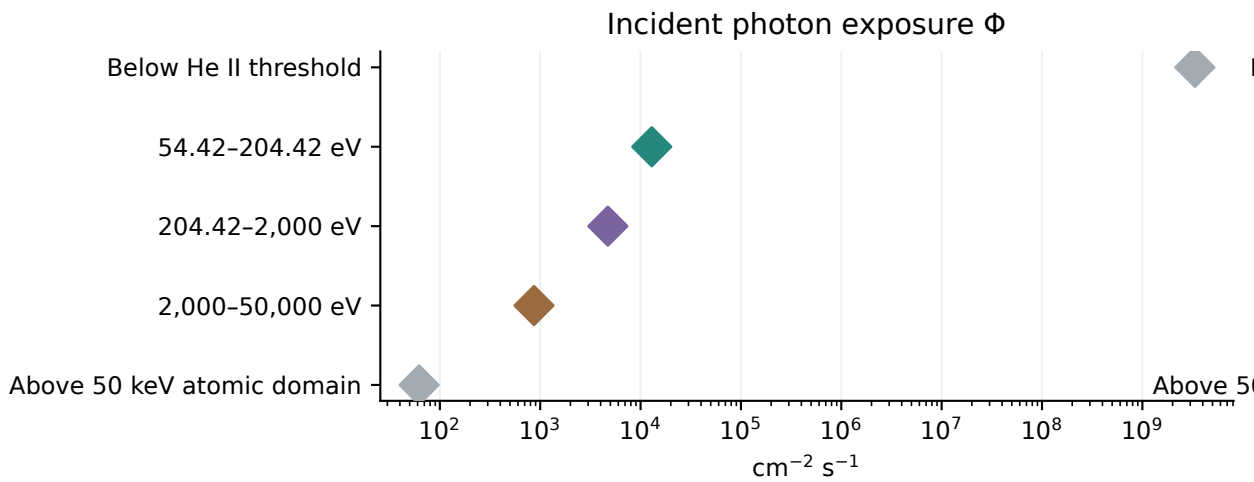


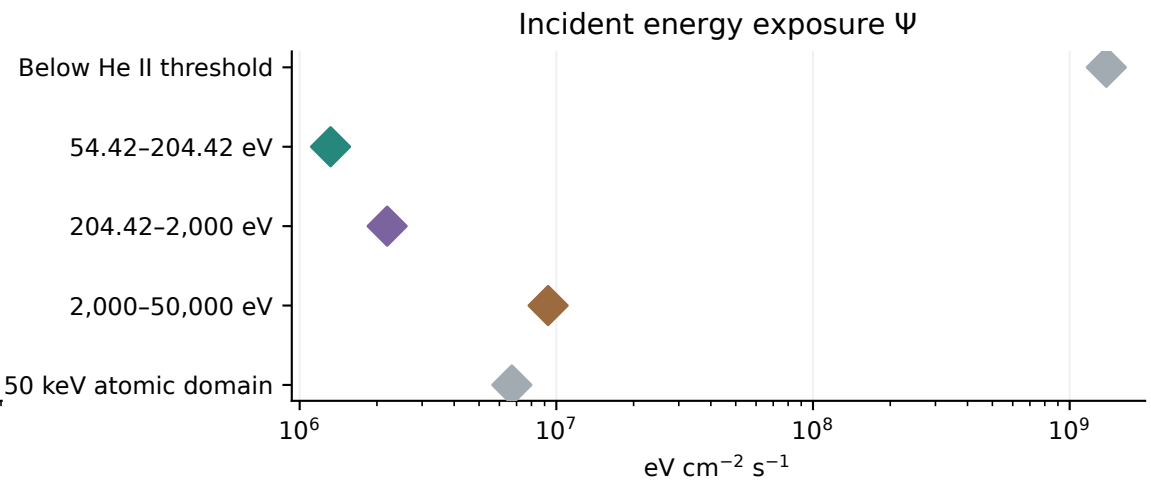












Below 54.42 eV: He II ionization is physically zero. Above 50 keV: radiation exposure known, atomic rate unknown.
Local spectral-operator comparison only; no gas history, heating deposition, Lumina cell, or observed reionization curve.